SEVA INNOVATION CHALLENGE 2026 | PROJECT PRESENTATION

BIS Procurement Assistant

Tender-to-standard discovery for procurement teams

What this prototype does

- Accepts tender PDF or typed requirements
- Extracts key product and specification fields
- Searches official BIS catalogue metadata
- Ranks up to 3 candidate standards with evidence

Discovery support only: review the official BIS record before making a procurement decision.

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The Problem

A practical software prototype for tender-to-standard discovery.

What is the problem?

Tender requirements often arrive as long PDFs or free text. Turning them into focused BIS catalogue searches takes time and familiarity with standards terminology.

Who is affected, and where?

Procurement teams and suppliers reviewing product tenders that reference Indian Standards, across public purchasing workflows.

Evidence

Project evidence: the app accepts PDF and manual-text input, extracts key requirement fields, and links candidate records to the official BIS search portal.

Current gaps

Catalogue search metadata does not include full standard clauses. A keyword match cannot establish compliance or certification eligibility.

Tender to BIS

One focused workflow for turning requirements into a shortlist of standards to review.

Better discovery can make standards review more structured. Confirm current scope and applicability with BIS.

3 Our Solution & Innovation

From tender input to an evidence-backed standards shortlist.

Our solution in brief

BIS Procurement Assistant accepts a tender PDF or typed requirements, extracts key parameters, searches BIS catalogue metadata, and returns up to three evidence-backed candidates with official search links.

How it works



What makes it new?

Workflow	Today	BIS Assist
Input	Read tender	PDF or text
Search	Manual terms	BIS metadata
Evidence	Hard to trace	Terms + links
Decision	Human review	Shortlist, verify

Key USP

USP: One workflow combines requirement extraction, official BIS metadata search, explainable keyword evidence, and verification links.

4 Feasibility & Implementation

Current build status, operating needs, and a proposed pilot plan.

Current stage: Working prototype (frontend + API) | Pilot not yet confirmed



Technology

React + Vite frontend; FastAPI API; SQLite; PyMuPDF text extraction; BM25 keyword ranking. Optional local SBERT is disabled unless configured.

Estimated cost

Not yet estimated. Assess hosting, security, support, and any model costs during pilot planning.

Timeline

Proposed plan: 0-3 months user testing; 3-6 months pilot and catalogue review; 6-12 months evaluate before scaling.

Partners

No government or industry pilot partner is confirmed in the project files. Identify a procurement team and BIS domain reviewer.

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Impact & Scalability

Impact targets will be set after real users test the prototype.

TBD

Pilot users to be served

set with pilot partner

TBD

Review time saved

measure in pilot

TBD

Tender teams supported

set with pilot partner

Social impact

A structured shortlist may help reviewers trace why standards were suggested and open the official BIS record for review.

Sustainability

Measure review time, shortlist usefulness, user trust, and correction rates before estimating savings or operating costs.

Scale-up plan

Pilot one procurement workflow; review search coverage and false matches; then expand categories using approved BIS data sources.

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Project Workstreams & Support Required

A software prototype is in place; next comes domain review and a controlled pilot.

Frontend

UI

React + Vite

Upload, forms, dashboard and results UI

Backend API

API

FastAPI + SQLite

Accounts, tender history and reports

Requirements

REQ

Rule-based extraction

Key fields from typed text or text PDFs

BIS search

BIS

Metadata + BM25

Evidence and official search links

Workstreams shown; verified member names/contact details were not in the project files.

Support we are seeking

- [x] Mentorship & technical guidance
- [x] Pilot with a government department
- [ ] Funding / grant support
- [ ] Incubation & workspace
- [ ] Testing, certification or IP support

Priority: BIS domain review, procurement-team pilot access, and guidance on authorized catalogue data and evaluation.

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